The University of Western Australia
SCHOOL OF MATHEMATICS & STATISTICS

AMO TRAINING SESSIONS

Australian Mathematics Olympiad, 2009 Trial Problems with Some Solutions and Some Hints

1. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x)f(y) - f(xy) = x + y$$

for all $x, y \in \mathbb{R}$.

Solution. We try a few values, in an attempt to find some defining properties for f. First we try putting $x = y = 0$. Then

$$\begin{aligned} f(0)f(0) - f(0) &= 0 \\ f(0)\big(f(0) - 1\big) &= 0 \\ \therefore f(0) &= 0 \text{ or } f(0) = 1. \end{aligned}$$

Putting $x = 0$, $y = 1$, we have

$$\begin{aligned} f(0)f(1) - f(0) &= 1 \\ f(0)\big(f(1) - 1\big) &= 1 \\ \therefore f(0) &\neq 0 \\ \therefore f(0) &= 1. \end{aligned}$$

Finally, leaving $x = x$ and substituting $y = 0$, we have

$$\begin{aligned} f(x)f(0) - f(0) &= x \\ f(x) - 1 &= x \\ \therefore f(x) &= x + 1. \end{aligned}$$

Checking, we have:

$$\begin{aligned} f(x)f(y) - f(xy) &= (x + 1)(y + 1) - (xy + 1) \\ &= x + y. \end{aligned}$$

Thus there is just the one function with the property, namely

$$f(x) = x + 1.$$

2. In a latest theory of particle physics there are three fundamental particles. When two particles of different types collide they are replaced by a particle of the third type. Two particles of the same type never collide. Prove that if an experiment begins with an equal number of particles of each type it cannot end with just one particle remaining.

Solution. We are done if we can find a property that is invariant over each step of the experiment. Observe that the numbers of two types of particles decreases by 1 and the other type increases by 1, so that the total number of particles decreases by 1 at each step. So the experiment proceeds for only a finite number of steps, terminating when there is only one type of particle remaining.

Since at each step, the numbers of each type of particles changes by 1, the parity (i.e. what it is modulo 2) of the numbers of each type of particle changes at each step. If (ℓ, m, n) represents that there are ℓ, m, n of the three types of particle, respectively, then the initial state is (N, N, N) for some $N \in \mathbb{N}$. Thus initially, the numbers (and hence the parities of those numbers) of each particle type are the same, and since all the particle types swap parity at each step, we have an invariant property, namely that: the numbers of particles of each type have the same parity.

At each step, the state parity toggles between (even, even, even) and (odd, odd, odd). Since $(0, 0, 1)$ (and any permutation) is of neither parity type, the experiment can never finish with just one particle.

3. A chord PQ with midpoint R is drawn in a circle with diameter AB. Perpendiculars PS and QT are dropped onto AB.

(i) Prove that $\triangle RST$ is isosceles.

(ii) Prove that $\triangle RST$ is equilateral if and only if $2PQ = AB$.

4. Find the smallest positive integer which has exactly 30 positive divisors.

Solution. Let $N \in \mathbb{N}$. If its prime decomposition is given by

$$N = \prod_{k=1}^{n} p_k^{e_k},\ p_k \text{ prime},\ e_k \in \mathbb{N},$$

then the number of (positive) divisors of N is given by

$$\prod_{k=1}^{n} (e_k + 1).$$

Now 30 itself has prime decomposition $2 \cdot 3 \cdot 5$, and so has 8 divisors, so that

$$30 = 1 \cdot 30 = 2 \cdot 15 = 3 \cdot 10 = 5 \cdot 6$$

must represent all ways 30 can be written as the product of two natural numbers. By writing 30 in the form $\prod_{k=1}^{n}(e_k + 1)$ we may infer that a number of form $\prod_{k=1}^{n} p_k^{e_k}$ has 30 divisors. Let p, q, r be distinct primes. Then

$$\begin{array}{lll}
\text{for } n = 1\text{: } 30 = 29 + 1 & \Longrightarrow \text{numbers } N \text{ of form: } N = p^{29} & \text{have 30 divisors;} \\
\text{for } n = 2\text{: } 30 = (1+1)(14+1) & \Longrightarrow \text{numbers } N \text{ of form: } N = pq^{14} & \text{have 30 divisors;} \\
\qquad\qquad 30 = (2+1)(9+1) & \Longrightarrow \text{numbers } N \text{ of form: } N = p^2q^9 & \text{have 30 divisors;} \\
\qquad\qquad 30 = (4+1)(5+1) & \Longrightarrow \text{numbers } N \text{ of form: } N = p^4q^5 & \text{have 30 divisors;} \\
\text{for } n = 3\text{: } 30 = (1+1)(2+1)(4+1) & \Longrightarrow \text{numbers } N \text{ of form: } N = pq^2r^4 & \text{have 30 divisors.}
\end{array}$$

The total of the powers of the last type is the least, from which we may guess that $N = 2^4 \cdot 3^2 \cdot 5 = 720$ is the least natural number possessing 30 divisors. We now prove this claim.

$$\begin{aligned}
N &= p^{29} \geq 2^{29} > 2^{10} > 1000 \\
N &= pq^{14} = (pq)q^{13} \geq (2 \cdot 3)2^{12} > 2^{15} > 1000 \\
N &= p^2q^9 = (pq)^2q^7 \geq (2 \cdot 3)^2 2^5 > 2^{11} > 1000 \\
N &= p^4q^5 = (pq)^4 q \geq (2 \cdot 3)^4 2 = 2^5 \cdot 3^4 > 1000 \\
N &= pq^2r^4 = (pqr)(qr)r^2 \geq (2 \cdot 3 \cdot 5)(2 \cdot 3)2 = 720
\end{aligned}$$

Thus we see that any natural number possessing 30 divisors, is at least 720, and 720 does possess 30 divisors. So indeed 720 is the least natural number possessing 30 divisors.

5. Let $a, b, c \in \mathbb{R}$ such that $0 < a, b, c \leq 1$. Prove that

$$\frac{a}{bc+1} + \frac{b}{ca+1} + \frac{c}{ab+1} \leq 2.$$

Solution. Assume $a, b, c \in \mathbb{R}$ such that $0 < a, b, c \leq 1$. Then

$$\begin{aligned}
1-a \geq 0,\ 1-b \geq 0,\ 1-c \geq 0,\ ab > 0& \\
\therefore (1-a)(1-b) + ab + (1-c) > 0& \\
1 - a - b + ab + ab + 1 - c > 0& \\
2(1+ab) &> a+b+c \\
1 + ab &> \frac{a+b+c}{2} > 0 \\
\therefore \frac{1}{1+ab} &< \frac{2}{a+b+c}
\end{aligned}$$

Similarly,

$$\begin{aligned}
\frac{1}{1+bc} &< \frac{2}{a+b+c} \\
\frac{1}{1+ca} &< \frac{2}{a+b+c} \\
\therefore \frac{a}{bc+1} + \frac{b}{ca+1} + \frac{c}{ab+1} &< \frac{2a}{a+b+c} + \frac{2b}{a+b+c} + \frac{2c}{a+b+c} \\
&= \frac{2(a+b+c)}{a+b+c} = 2.
\end{aligned}$$

Observe that we proved something slightly stronger than required. Also, observe that the restriction that a, b, c be 0 is unnecessary, and if $a = b = 1$ and $c = 0$ then the statement is true with equality.

6. Number plates in a certain country have 6 digits ranging from 000000 to 999999. How many number plates can be issued such that any 2 number plates differ in at least 2 places?

7. For any positive integer n, let $T(n)$ be the number of positive divisors of n. Find all n for which the sequence

$$n, T(n), T\big(T(n)\big), \ldots$$

does not contain a perfect square.

Solution. Consider the following cases of n.

Case (1): $n = 1$. Then the sequence is:

$$1, 1, \ldots$$

all of whose terms are 1, a square.

Case (2): $n = p$, prime. A prime number has two divisors: 1 and p. So the sequence is:

$$p, 2, 2, \ldots$$

none of whose terms is a square.

Case (3): n is composite. If $m \in \mathbb{Z}$ such that $m > 2$ then

$$2 \leq T(m) < m$$

since $1 \mid m$ and $m \mid m$ ($\implies T(m) \geq 2$), and $(m-1, m) = 1$ ($\implies$ the set of divisors of m is a subset of $\{1, 2, \ldots, m\} \setminus \{m-1\}$ for $m > 2$, i.e. $T(m) < m$). Thus the sequence strictly decreases as long as its terms m are greater than 2. Hence, eventually one of the terms of the sequence is 2. But $T(2) = 2$, so that the sequence is constant thereafter. Thus the sequence is of form

$$n, T(n), \ldots, x, T(x), 2, 2, 2, 2, \ldots$$

for some x such that $T(x) > 2$, but such that $T\big(T(x)\big) = 2$. Now, $T\big(T(x)\big) = 2$ implies that $T(x)$ is prime, but $T(x) \neq 2$ so that it is an *odd* prime. But the only natural numbers x that have an odd number of divisors are squares. So x is a square. Thus if n is composite then the sequence necessarily contains a square.

Thus the numbers $n \in \mathbb{N}$ for which the given sequence does not have a square are precisely the prime numbers.

8. A quadrilateral $ABCD$ is inscribed in a circle so that AB is a diameter. Suppose that AB and CD intersect at I, AD and BC intersect at J, AC and BD intersect at K and let N be a point on AB. Prove that $IK \perp JN$ if and only if N is the midpoint of AB.